



2807 West County Road 75
Monticello, MN 55362

September 25, 2025

L-MT-25-029
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Subsequent Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant Licensee Event Report 2025-004-00

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-263/2025-004-00 per 10 CFR 50.73(a)(2)(iv)(A), 10 CFR 50.73(a)(2)(v)(C), and 10 CFR 50.73(a)(2)(v)(D).

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Michelle Neal'.

Michelle Neal
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-263/2025-004-00**

3 pages follow

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Monticello Nuclear Generating Plant	☒ 050	2. Docket Number -263	3. Page 1 OF 3
	☐ 052		

4. Title
Rainwater Intrusion Caused Failure of Reactor Building Ventilation and Rise in Steam Chase Temperatures

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
07	28	2025	2025	004	00	09	25	2025	☐ 050	
									Facility Name	Docket Number
									☐ 052	

9. Operating Mode Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Nathan Fedora, Senior Nuclear Regulatory Engineer	Phone Number (Include area code) 612-342-8971
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	VL	FAN	R165	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 22:00 on July 28, 2025, with Monticello Nuclear Generating Plant (MNGP) operating at 100 percent power, secondary containment was declared inoperable due to a Reactor Building Ventilation System failure which caused a degradation of Reactor Building differential pressure that exceeded the Technical Specification limits. This Ventilation System failure led to elevated temperatures in the Steam Chase. Operators began to lower Reactor power to attempt to slow the rising Steam Chase temperature level. At 23:24 on July 28, 2025, with MNGP now operating at 41 percent power, the Reactor was manually scrammed due to Steam Chase temperatures reaching the procedural limit. Operators responded to the trip and stabilized the plant in accordance with approved procedures and safely placed the plant in Mode 3. All systems responded normally post trip. The Ventilation System failure was caused by rainwater intrusion through a leak in the plenum room roof that made contact with and then tripped energized Ventilation System components. The entire plenum room roof is in the process of being replaced. Immediate actions were taken to prevent rainwater intrusion from impacting Ventilation System components while the roof is being replaced.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Monticello Nuclear Generating Plant	☒ 050	2. DOCKET NUMBER -263	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 004	REV NO. 00

NARRATIVE

EVENT DESCRIPTION AND ANALYSIS

On the evening of July 28, 2025, Monticello Nuclear Generating Plant (MNGP) was operating at 100 percent power when a thunderstorm passed over the area. As the storm passed, rainwater began dripping through the roofing material into the plenum room which is located between the Reactor Building and Turbine Building. The water ran down the wall above cabinet N1131, V-EF-24 LOCAL CONTACTOR. At about 20:58 on July 28, 2025, this leak in the plenum roof led to an electrical short of a contactor for motor V-EF-24, REACTOR BUILDING EXHAUST FAN (Reliance Electric, model BFC SO7827-1) [VL]. This short caused a fuse in a General Electric 7700 line breaker B1131, V-EF-24A,B, REACTOR BUILDING MAIN EXHAUST FANS 480 VAC, to open resulting in a loss of power to both Reactor Building main exhaust fans. Then, V-AC-10A and V-AC-10B, REACTOR BUILDING MAIN SUPPLY HVAC UNITS, [VA] tripped as designed with the V-EF-24 flow interlock.

With the failure of Reactor Building Exhaust Ventilation and then the subsequent trip of the Reactor Building Supply Ventilation, this resulted in a degradation of the Reactor Building differential pressure. At 22:00, on July 28, 2025, with MNGP still operating at 100 percent power, the differential pressure exceeded the Technical Specifications limit resulting in Secondary Containment being declared as inoperable.

With the failure of the Ventilation System, Steam Chase temperatures started to rise. Operators began to lower Reactor power to attempt to slow the rising Steam Chase temperature level.

At 23:24 on July 28, 2025, with MNGP now operating at a reduced power of 41 percent, the Operators inserted a manual scram of the Reactor due to the Steam Chase temperatures reaching the procedural limit of 180 degrees Fahrenheit per the criteria in the Ventilation System procedure.

This event is reportable per 10 CFR 50.73(a)(2)(v)(C) and 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material; and mitigate the consequences of an accident, and per 10 CFR 50.73(a)(2)(iv)(A) due to actuation of the Reactor Protection System.

ASSESSMENT OF SAFETY CONSEQUENCES

The scram was not complex, with all systems responding normally post scram. Reactor water level was maintained via the Feedwater System. Decay heat was removed by discharging steam to the main condenser using the turbine bypass valves.

The health and safety of the public and site personnel were not impacted during this event.

CAUSE OF THE EVENT

The Ventilation System failure was caused by rainwater intrusion through a leak in the plenum room roof that then made contact with and tripped energized Ventilation System components



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				2025	004	00

NARRATIVE

CORRECTIVE ACTIONS

Immediately after the event, sealant was applied to electrical cabinets in the plenum room to prevent rainwater intrusion from impacting the Ventilation System components.

To reduce the likelihood of this event happening again, replacement of the entire plenum roof is currently in progress.

MNGP was placed back online at 08:30 on August 2, 2025.

PREVIOUS SIMILAR EVENTS

There were no previous similar events in the past three years.

ADDITIONAL INFORMATION

All times are in Central Daylight Time.

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].